# Scale, proportions, circles, and review

25 questions, including 2 challenges. Write your work and answers on separate paper.

1. $12.2 \times 10$

2. $1200 \div 100$

3. Double 12.5

4. Half of 72

5. Compare $\frac{2}{4}$ and $\frac{2}{5}$.

6. Yellow : purple is 2 : 5. There are 35 beads. Find both counts.

7. Find 140% of 30 points.

8. Which is the reciprocal of $2\frac{1}{2}$: $\frac{2}{5}$ or $\frac{5}{2}$? Explain using a product of 1.

9. Find 2.45 + 4.078.

10. Solve $1.5x = 4.80$.

11. Find the surface area of a closed 1 by 4 by 5 cm box.

12. Write an inequality for x that means less than 5. Is the boundary included?

13. A: 3, 4, 5. B: 1, 4, 7. Compare their means and ranges.

14. A circle has diameter 3 cm. Find circumference exactly and using 3.14.

15. A garden is 3 cm by 4 cm on a map where 1 cm means 10 m. Redraw it where 1 cm means 20 m. Find the new drawing dimensions and the actual garden area.

16. A rectangle 7 cm by 2 cm is scaled by 2. Find new dimensions and area.

17. A drawing has area 14 square cm. Every length is scaled by $\frac{1}{2}$. Find the new area and explain the factor.

18. A map uses 1 cm for 7 km. A route is 2.5 cm on the map. Find actual distance.

19. A proportional table includes (x, y) = (2, 14). Find k, and find y when x = 5.

20. A rental costs $7 to start plus $2 per hour. Write cost c for h hours. Is the relationship proportional?

21. A pump fills 7 liters in 2 minutes. At a constant rate, write L after t minutes and find the amount after 4 minutes.

22. A circle has diameter 14 cm. Find circumference in terms of π and using 3.14.

23. A circle has radius 2 cm. Find area in terms of π and using 3.14.

## CHALLENGE / Explain your plan and check the result

C1. A circular running track has diameter 40 m. Using 3.14, find the distance of 3 full laps plus half a lap. A runner covers it in 7 minutes. Find the average speed in meters per minute.

C2. Four orders cost $18, $22, $26, and $34. Find the mean. A fifth order is added so the new mean is $27. Find its cost and the new median. Explain why the new mean and median differ.